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Neural Trajectories of Conceptually Related Events

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eLife Assessment

This manuscript investigates the extremely interesting and **important** claim that the human hippocampus represents interactions with multiple social interaction partners on two relatively abstract social dimensions - and that this ability correlates with the social network size of the participant. This research potentially demonstrates the intricate role of the hippocampus in navigating our social world. While most of the results are **solid**, the paper requires some further clarification.

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Abstract

In a series of conceptually related episodes, meaning arises from the link between these events rather than from each event individually. How does the brain keep track of conceptually related sequences of events (i.e., conceptual trajectories)? In a particular kind of conceptual trajectory—a social relationship—meaning arises from a specific sequence of interactions. To test whether such abstract sequences are neurally tracked, we had participants complete a naturalistic narrative-based social interaction game, during functional magnetic resonance imaging. We modeled the simulated relationships as trajectories through an abstract affiliation and power space. Using two independent samples ($n = 50$), we found evidence of both the underlying social dimensions of affiliation and power and the evolving social relationships themselves as being tracked by the hippocampus. These results suggest that our evolving relationships with others, despite being composed of isolated events that occur independently across time and space (i.e., spatially and temporally non-consecutive events), but are nevertheless related conceptually, are represented in trajectory-like neural patterns in the human hippocampus.

Introduction

When you are in a certain place, say a hotel lobby, the meaning of that place (i.e., your location in state space, or latent state) only makes sense in the context of what led you there. A sequence of observations is necessary to disambiguate the current state, as a single observation is insufficient: checking in and checking out of a hotel may look identical yet they predict very different futures. This also applies to conceptual spaces, where sensory information is fully abstracted out. A prototypical case of conceptual space is social space, where the state of a social relationship evolves over interactions along the latent dimensions of power and affiliation¹. For example, a conversation between two people in that hotel lobby can mean different things if they are old friends, lifelong enemies, or strangers. The prior relationship state—itsself a summary of conceptually connected events from the relationship history—helps determine both the current and future relationship states. But while neural representations of latent states and paths in

physical space have been identified^{2,3}, as have location-like representations in conceptual spaces^{4,5}, including social^{1,6}, it is unknown whether the brain dynamically tracks locations as a sequence of states in conceptual space.

Where and how could relationship trajectories through social space be represented in the human brain? The hippocampus is a likely candidate. Functional magnetic resonance imaging (fMRI) studies have shown that the hippocampus tracks latent states across domains, from physical locations⁷ to concepts^{8,9} to abstract relations between self and other^{6,10,11}. Sequences of neural states in the hippocampus may also be connected like trajectories. Evidence consistent with this idea showed that post-task fMRI activity patterns reactivated sequentially¹², reflecting previously learned sequences. An intriguing possibility is that the hippocampus maps sequences of social interactions in relationships onto a neural manifold, tracking them like trajectories in an abstract social space.

To examine this, we used representational similarity analyses¹³ to test whether hippocampal fMRI patterns show structures that are specific to different relationships' trajectories, but that share the same representational geometry. Participants completed a naturalistic and narrative-based social interaction task⁶, where they interact and form relationships with fictional people in a new social network. The interactions are defined by choices with latent affiliation and power content, and the relationships with the characters evolve on these dimensions as the task progresses (see [Figure 1](#)). We tested the following hypotheses about hippocampal representations: First, we expected the hippocampus to represent the social interactions on affiliation (e.g., cooperation) and power (e.g., hierarchy) dimensions, a compressed format that allows generalization across social situations. Across the different characters and interactions, hippocampal patterns should be more similar for interactions of the same dimension than interactions of a different dimension. But while these dimensions are general, the individual relationships are specific: each relationship should have hippocampal representations that reflect the trial-by-trial behavioral trajectories in a shared social space. Our results support these hypotheses and suggest that the hippocampus tracks sequences of relationship-specific social interactions like trajectories on a common manifold. These results provide evidence for hippocampal linking of conceptually related events, even when separated in time and context, into coherent narratives.

Results

Behavioral geometry is consistent with social mapping

Before running the fMRI analyses, we asked whether participants' choices were consistent with a map-like representation in affiliation and power space. We tested two main behavioral assumptions: (1) behavior is independent between participants (i.e., not a task artifact shared across participants) and (2) behavioral locations align with participants' self-reported placements of the characters ([Figure 1](#)). Below, we discuss each of these.

Participants' behavior is idiosyncratic

If the task structure determined behavior, participants would have made similar choices and traversed similar paths through social space. However, each participant's set of choices was unique (i.e., never identical between participants). The character specific trajectories were also largely unique: out of the character specific choices (250 for 50 participants with 5 characters each), 96.8% (242/250) were unique. Moreover, the behavioral trajectories approximately occupied the entire space across both samples. As such, participant behavior was idiosyncratic and unlikely to simply reflect task structure.

Behavioral locations are related to self-reported locations

To test if the participants subjectively represented the social locations implied by their behavioral choices, we asked whether self-reported (i.e., explicit) placements of the characters into the social space align with the behavioral (i.e., implicit) locations from the task. After the task we had the Validation sample participants (n = 32) place the characters into a two-dimensional (2D) affiliation

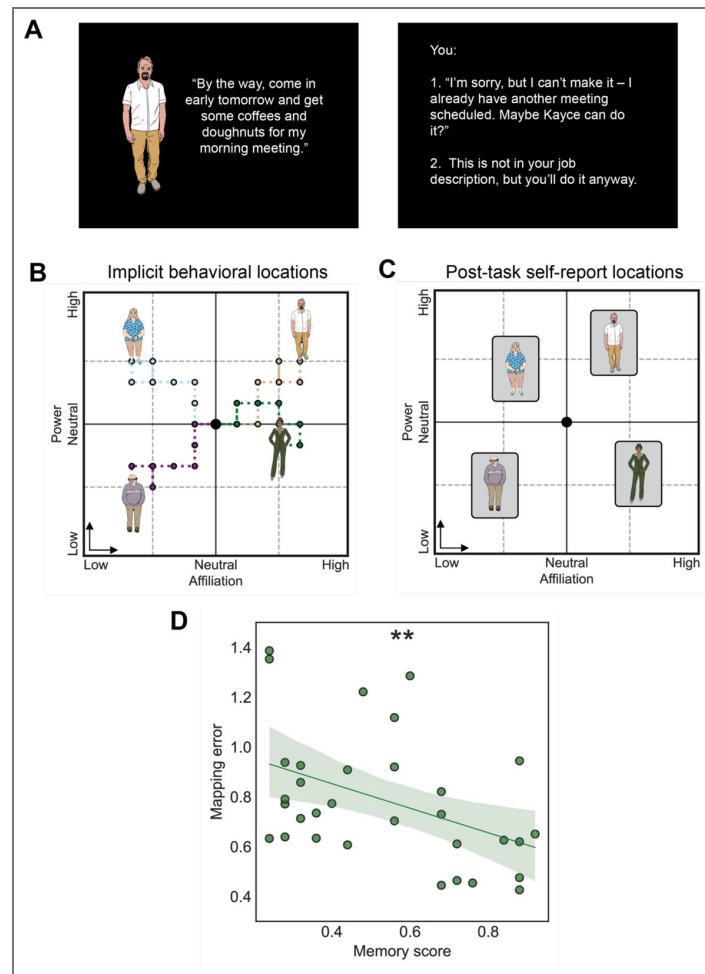


Figure 1. Social interaction sequences form relationship trajectories along abstract dimensions of affiliation and power.

(A) An example of a power interaction. Participants read text that describes the narrative and on decision trials choose between two options. Based on their choice, the character moves -1 or $+1$ along the active dimension (power or, on other trials, affiliation). (B) The participant forms relationships with different characters through sequences of interactions in the narrative. The decisions the participant makes in the implicit affiliation and power interactions change the character's location in social space, forming a relationship trajectory (participants are unaware of the dimensions). Participants interact with 6 characters: 5 each with 6 affiliation and 6 power trials and 1 with 3 neutral trials. 4 characters are shown for illustration. (C) Schematic of post-task self-report placements. The mapping error between the behavioral and subjective placements was calculated as the average character-wise Euclidean distance between the locations; this was significantly smaller than expected by chance. (D) Mapping error was negatively correlated with task memory, suggesting the subjective maps depend on memory. 95% confidence intervals for regression line are indicated by the shaded region and p-value significance is indicated by asterisks: $* < 0.05$. Validation sample only ($n = 32$).

and power space based on their perception of their relationships with the characters. We calculated the participant-specific average distance between the characters' behavioral and the self-reported locations (i.e., "mapping error") and compared them to participant-specific permutation generated distances to get z-scores. A one-sided t-test with these z-scores showed that the distances were smaller than participant-specific permutation-based chance ($\beta = -0.563$, $CI_{95} = [-0.935, -0.191]$, $t_{31} = -3.09$, $p < 0.005$). We also found a predicted negative correlation between mapping error and task memory ($r = -0.44$, $CI_{95} = [-0.767, -0.108]$, $t_{31} = -2.71$, left-tailed $p < 0.01$), consistent with participants using their memory of the relationships to place the characters in social space. Together these results suggest that affiliation and power locations reflect a subjectively accessible map of the relationships, supporting the internal validity of the task.

The hippocampus represents the dimensions of affiliation and power during social interactions

After validating our main behavioral assumptions, we turned to the fMRI data. We hypothesized that the social interactions are represented abstractly along the affiliation and power dimensions in the hippocampus, such that they generalize across characters and interactions. If true, decision trial pairs of the same dimension should have more correlated patterns than decision trial pairs of different dimensions: the average pattern correlations should be higher for same dimension versus different dimension decision trial pairs. We used a representational similarity analysis (RSA) searchlight to test this prediction. For each participant, we computed the correlation differences between the same and different dimension trial pairs using a moving searchlight. Then, in the group-level model, we controlled for sample and the average reaction time difference between affiliation and power decisions and tested for a positive effect of correlation difference.

We first tested this in the bilateral hippocampus. There was a significant cluster, with the peak voxel in the anterior hippocampus (peak voxel MNI x/y/z = -34/-17/-16, cluster extent = 724 voxels), supporting our a priori hypothesis that the hippocampus represents social situations along the dimensions of affiliation and power (see Figure 2 [↗](#)). Importantly, character identity and temporal distance (i.e., the time between trials) are experimentally balanced across affiliation and power decision trials, and cannot explain these effects.

After testing our hippocampus prediction, we ran a whole-brain exploratory analysis. This showed additional significant clusters, suggesting a distributed network represents these dimensions: the bilateral angular gyrus (right: peak voxel MNI x/y/z = 44/-57/35, cluster extent = 1824 voxels; left: peak voxel MNI x/y/z = -52/-66/29, cluster extent = 1303 voxels) and the bilateral middle temporal gyrus (right: peak voxel MNI x/y/z = 51/-7/-32, cluster extent = 327 voxels; left: peak voxel MNI x/y/z = -54/-7/-26, cluster extent = 574 voxels) Both the angular gyrus, a region situated within the temporo-parietal junction, and the middle temporal gyrus have been implicated in social cognition (e.g., theory of mind)^{14,15}.

The hippocampus represents affiliation and power locations within relationships

We have shown that the left hippocampus represents the affiliation and power trials differently, consistent with an abstract dimensional representation. Does it also represent the changing social coordinates of each character? To test this, we multiple-regression RSA searchlight to test whether left hippocampus patterns represent the characters' changing social locations across interactions (see Figure 3 [↗](#)). We restricted the distances to those from trial pairs from the same character and standardized the distances within character (see Figure 3B-D [↗](#)). We controlled for temporal distance to ensure the effect was not explainable by the time between trials, and for whether the trials shared the same underlying dimension (affiliation or power; see **Location similarity searchlight analyses** for more details). At the group level, we controlled for sample and the average reaction time difference between affiliation and power trials. Using the same testing logic as the dimensionality similarity analysis, we first tested our hypothesis in the bilateral hippocampus and found widespread effects in both the left (peak voxel MNI x/y/z = -35/-22/-15,

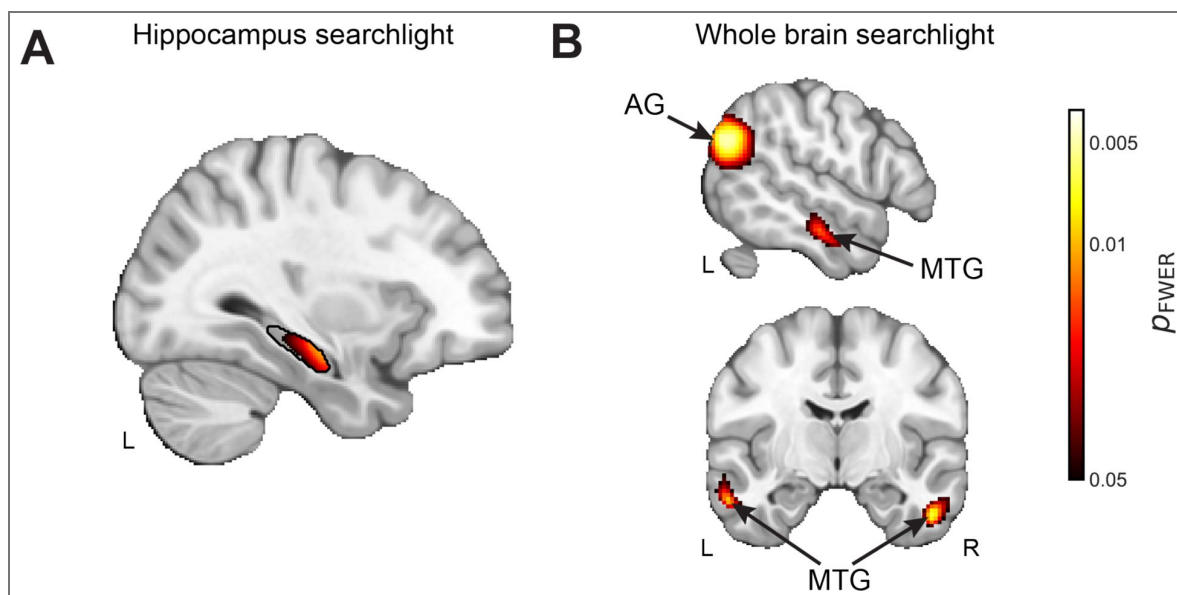


Figure 2. The left hippocampus represents affiliation and power decision trials differently.

Decision trials of the same dimension had more similar neural patterns than trials of different dimensions. A) An analysis restricted to an anatomically defined hippocampus showed a significant cluster in the left hemisphere. B) An exploratory whole-brain analysis showed additional clusters in the bilateral middle temporal gyrus (MTG) and angular gyrus (AG). Participants were pooled across samples ($n = 50$); sample included as covariate. Permutation-derived $p_{FWER} < 0.05$.

cluster extent = 1470 voxels) and right (peak voxel MNI x/y/z = 37/-19/-14, cluster extent = 1953 voxels) hemispheres. The whole-brain searchlight analysis revealed additional clusters in the left putamen (-27/-3/14, cluster extent = 131 voxels) and left posterior cingulate cortex (-10/-28/41, cluster extent = 304 voxels).

We then asked a second, complementary question: does the hippocampus represent all interactions, across characters, within a shared map? To test for this map-like structure, we repeated the analysis but now included all trial pairs, z-scoring distances globally rather than within character (Figure 3E-F). The remainder of the procedure followed the same logic as the preceding analysis. The hippocampus analysis revealed an extensive right hippocampal cluster (27/-27/-14, cluster extent = 1667 voxels). The whole-brain analysis did not show any significant clusters.

Discussion

Here, we study a case of conceptual trajectories, which are sequences of abstractly, rather than physically, related events, and whose order is meaningful rather than arbitrary or physically determined (spatial or temporal sequences). We show that by connecting conceptually related interactions, the hippocampus represents evolving social relationships as sequences of neural states.

We first show that left hippocampal patterns are more similar in social interactions with the same underlying dimension (i.e., affiliation compared to affiliation, and power compared to power) than interactions with different dimensions (i.e., affiliation to power), consistent with abstract affiliation and power representations. Then, we show that the hippocampus tracks the changing social locations (affiliation and power coordinates), above and beyond the effects of dimension or time; the hippocampus seemed to reflect both the changing within-character locations, tracking their locations over time, and locations across characters, as if in a shared map. Thus, these results suggest that the hippocampus does not just encode static character-related representations but rather tracks relationship changes in terms of underlying affiliation and power.

Previous work has shown that the hippocampus tracks temporally sequential events, such as trajectories in physical space (i.e., traversing sequentially connected locations¹⁶) as well as sequences of arbitrary information units, such as ordered lists of words or scenes (e.g.,¹⁷). Social relationships, however, are *conceptual* sequences: the events that constitute a relationship are spatially and temporally distant but conceptually linked, constituting a trajectory in abstract social space. Stacking the neural representations of these interactions together, then, would constitute the neural trajectory of the relationship through social space.

Why would the hippocampus encode abstract sequences with a neural trajectory? We cannot answer this question directly with fMRI, but there is a plausible account from the neurophysiological literature. Hippocampal neurons are known to track the physical locations of oneself¹⁸ and others¹⁹, as well as social identity²⁰ and non-spatial sensory information²¹, suggesting the hippocampus may compute locations in abstract spaces—including social space. Moreover, the activity of hippocampal cell populations is ordered with respect to task dimensions, such that population-level patterns track locations in task space^{21,22}. Because hippocampal firing fields overlap to cover the task space²³, trajectories through task space should activate sequences of correlated hippocampal patterns, forming a neural trajectory that reflects the behavioral trajectory. This activity may be intrinsically low-dimensional: many hippocampal cells co-activate to encode locations in task space, restricting the possible activity patterns to a low-dimensional subspace²⁴. Many possible neural states in theory become many fewer in practice, as neuronal correlations restrict intrinsic activity to low-dimensional manifolds. In the case of conceptually related events that are separated in time, space, and context—as in the case of episodes of social interactions over a lifetime—it is possible that at each interaction, the hippocampus retrieves a representation of the last interaction's social location and then infers how the current interaction changes the relationship's location.

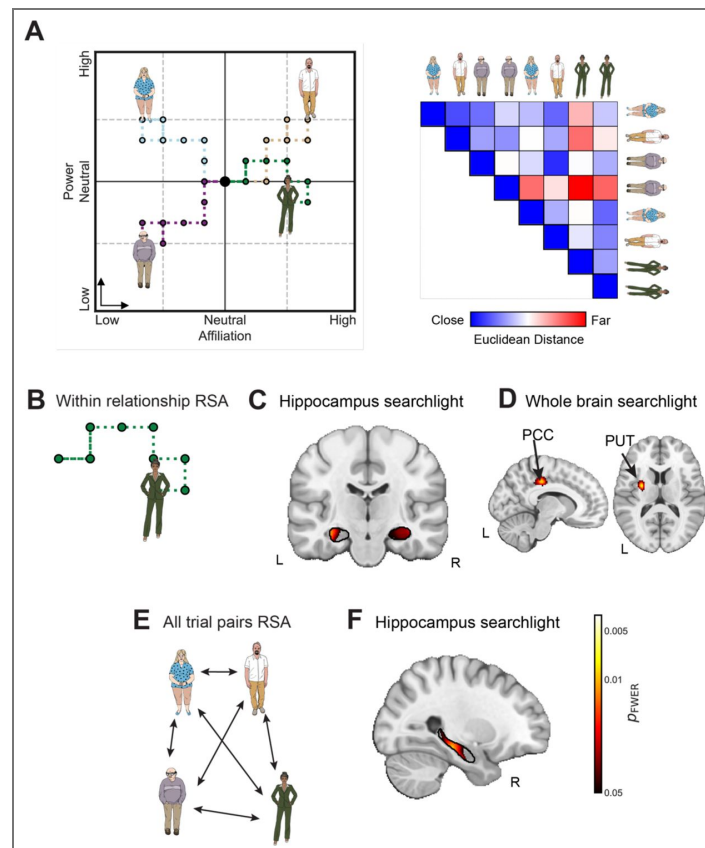


Figure 3. The hippocampus represents dynamically changing abstract social locations.

A) We used representational similarity analyses (RSA) to study location-like effects. We used the behavioral locations implied by the participants' choices to compute behavioral distance matrices. The distance matrix shown is a small, toy schematic to represent the logic; there were 60 decision trials in the task. We ran regressions and included control distances for task- and time-related variables to isolate the contribution of abstract location-related distances. B) We ran a within-character version of the RSA, where distances were within-character, to test for the brain dynamically tracking locations over relationship-specific social interactions. C) The anatomically defined left and right hippocampi both showed within-character location-like effects, in a region-of-interest analysis. The left shows effects from the mid- to posterior hippocampus, whereas the right was mainly in mid-hippocampus. D) An exploratory whole-brain analysis also showed clusters in the posterior cingulate cortex (PCC) and putamen (PUT), as well as a small cluster in the right inferior temporal cortex (not shown). E) We ran an additional RSA, including all trial pairs, to test for evidence of a shared map-like representation. F) In a hippocampal specific analysis, there was an effect in the mid-to-posterior right hippocampus. In these analyses, participants were pooled across samples ($n = 50$); sample included as covariate. Permutation-derived $p_{\text{FWE}} < 0.05$.

The approach used in this study can be applied to similar questions in non-social domains. For example, context-related effects^{2,5} may also be explained in a similar manner: memories of different episodes that share a higher-order context could have correlated patterns of activity, creating trajectories through a conceptual space. The structure of the manifold that these trajectories evolve along may hint at its underlying dynamics, allowing the formation and testing of novel computational hypotheses.

In summary, we provide evidence that the hippocampus represents social relationships with ordered sequences of neural patterns. Each relationship had its own sequence of states, and all relationships were embedded in the same manifold. These results put forward a novel way to look at representations of all kinds of evolving latent relationships, physical to social.

Methods

Independent samples

To ensure the effects we find are robust, we tested our hypotheses in two independent samples, called the “Initial” and “Validation” samples. The Initial sample was collected in a previous study³ and included 18 of 21 participants (8 female), after excluding 1 for psychiatric evaluation (Psychiatric Diagnostic Screening Questionnaire) and 2 for DICOM corruption during archiving. The Validation sample included 32 of 39 participants (17 female), after excluding 2 for poor post-task character memory ($\leq$ chance [20%] for 5 main characters) and 5 for high motion (mean framewise displacement $\geq$ 0.3mm). The two samples did not differ in terms of participant sex ($\chi^2 = 0.45$, $p = 0.5$) but were different in age (in years; Initial: mean (M) = 29.33, standard deviation (SD) = 3.42; Validation: $M = 44.42$, $SD = 9.69$; Welch’s unequal variances t-test $p < 0.001$).

The Institutional Review Board of the Icahn School of Medicine at Mount Sinai approved the experimental protocols for both samples. All participants provided written informed consent and were paid for their participation.

Social Navigation Task

Task description

The Social Navigation Task is a narrative-based social interaction game. Pre-task instructions are minimal: participants are told they will complete a task where they interact with different fictional characters, and to just behave as they naturally would. Importantly, the participants are never told or otherwise taught about the affiliation and power dimensions underlying the interactions. From the point-of-view of the participant, they are simply interacting with characters in a narrative and the affiliation and power dimensions are fully latent (i.e., unobserved or implicit).

There were 6 characters; their gender and racial presentations were counterbalanced. Gender was flipped across two versions (three men and three women), and in the Validation sample, the characters’ skin color was also counterbalanced with lighter- and darker-skinned characters variants of the same base images. The task version used for the Initial sample had approximately the same text and options as the Validation sample’s version (with small modifications), but with more cartoon-like character images⁶.

The narrative begins with the participant being told that they have just moved to a new town, and they need to find a job and a place to live. It unfolds over various social interactions that vary in their details and then culminates in the participant having successfully found employment and housing. There are two types of trials: “narrative” trials where background information is provided or characters talk or take actions (a total of 154 trials), and “decision” trials where the participant makes decisions in one-on-one interactions with a character that can change the relationship with that character (a total of 63 trials). On each decision, participants used a button response box to select between the two options. The options (1 or 2, assigned to the index and middle fingers) choice directions (± 1 arbitrary unit on the current dimension) were counterbalanced.

The sequence of trials, including both narrative and decision trials, were fixed across participants; all that differs are the choices that the participants make. Narrative trials varied in duration, depending on the content (range 2-10 seconds), but were identical across participants. Decision trials always lasted 12 seconds, with two options presented until the participant made a choice, after which a blank screen was presented for the remainder of the duration. All six characters' decision trials are interleaved with one another, and with the narrative slides. On average, after a decision trial for a given character, participants view ~11 narrative slides and complete ~3 decisions for other characters before returning to another decision with the same character, such that each character's choices are separated by an average of ~20 seconds (ranging from 12 seconds to 10 min).

Affiliation and power decisions

Decision trials were categorized as either affiliation or power decisions. Each decision trial presented two options, one increasing (+1) and one decreasing (−1) the character's position along the relevant dimension. Five main characters each had 6 affiliation and 6 power decisions, and a sixth neutral character had 3 neutral decisions (i.e., where the choices were inconsequential; not used for analysis). Affiliation decisions were defined as decisions whether to share physical touch, physical space or information (e.g., to share their thoughts on a topic). Power decisions were defined as decisions to issue or submit to a directive/command, or otherwise exert or give control. Examples of both trial types are shown in [Table 1](#).

To ensure that any observed neural or behavioral differences were not confounded by trivial features of the text, we tested for differences between the affiliation and power trials (where the two options are concatenated). There were no differences in word count (affiliation average = 26.6, power average = 25.6; t-test $p = 0.56$). The text's sentiment also did not differ between these trial types (t-test $p = 0.72$), as quantified by comparing sentiment compound scores (from most negative, −1, to most positive, +1), using a Large Language Model (LLM) specialized for sentiment analysis.

Our framework assumes that affiliation and power trials differ in their semantic content—that is, in the conceptual meaning of the text, beyond word count or sentiment. To test this assumption, we used an LLM-based semantic embedding analysis. Each decision trial was embedded into a semantic vector. We then measured the cosine similarity between pairs of trials and calculated the difference between average within-dimension similarity (affiliation-affiliation and power-power comparisons) and average between-dimension similarity (affiliation-power comparisons) and assessed its statistical significance with permutation testing (1,000 shuffles of trial labels). As expected, decision trials of the same dimension were more similar to each other than trials of different dimension, across multiple LLMs (OpenAI's text-embedding-3-small: similarity difference = 0.041, $p < 0.001$; all-MiniLM-L12-v2: similarity difference = 0.032, $p < 0.001$).

Post-task measures

Task memory

To validate that the participants attended to the task, participants answered 30 memory questions about the characters after the task was completed (outside the scanner). In each question, the options were 5 of the 6 characters (including the neutral): each character was presented as an option in 25 questions and was the correct answer in 5 of those questions.

Self-reported character placement

Participants may form a subjective representation of the characters' 2D social locations. To probe this, the participants completed a character placement task outside of the scanner after the task (Validation sample only). The participants were instructed to drag-and-drop colored dots representing each of the 6 characters onto locations in a 2D affiliation and power space, according to the participant's perceptions of their relationships with the characters. The placements were relative to the participant's theoretical point-of-view, which was represented by a red dot placed at the max affiliation and neutral/middle power values.

Table 1. Examples of affiliation and power interaction decisions.

Several examples of affiliation and power interactions are shown. The slide preceding the choice trial is shown to give the context for the interaction, along with the decision that would increase and the decision that would decrease the location along the relevant dimension.

Decision dimension	Previous slide context		Increase option	Decrease option
Affiliation	Chris goes in for a hug.		You hug him for a long moment.	You shake his hand instead.
	Affiliation	Anthony is taking the elevator downstairs with you.	You: "Looks like you're the real person running this place - everybody really relies on you!"	You check your email on your phone: "Is it always that hectic in here?"
	Affiliation	Mrs. Newcomb gets a phone call. Something is wrong; she seems very worried.	You feel concerned: "Everything alright, Jane? What is it?"	You don't want to get involved: "I'll give you privacy. I think we're done, anyway."
	Power	Maya: "It doesn't matter if you're late. We're grabbing coffee."	You: "I can't - gotta go. Maybe tomorrow."	You really don't want to be late but reluctantly agree to grab coffee.
	Power	Kayce is approaching the car. Chris gestures for you to move over to the middle seat.	You wait for Chris to take the middle seat.	You immediately move over to the middle seat.
	Power	Anthony: "By the way, I have a lead on a place. It's a condo with two separate units."	You: "Get me the contract and floor plan and tell them you need until tomorrow."	You: "Let me know when you have more information."

Behavioral analysis

Behavioral modeling

Each character started the task at the neutral origin (0, 0) of the social space. With each participant choice, the current character's coordinates were implicitly updated in the positive or negative direction along the current interaction dimension (i.e., each decision moved the character ± 1 arbitrary unit along affiliation or power). For any given decision trial (t), the current character's (c) affiliation and power coordinates are the cumulative sums of the trial-wise affiliation and power decisions:

$$\text{Affiliation}_{c,t} = \sum_{i=1}^t \text{affiliation choice}_{c,i}$$

$$\text{Power}_{c,t} = \sum_{i=1}^t \text{power choice}_{c,i}$$

where choice value is -1 or $+1$ if the trial is an interaction of that dimension, and 0 if not. Thus, each participant's sequence of interaction choices create a relationship trajectory through the latent affiliation and power space (see [Figure 1](#) .

Self-reported character placement analysis

We modeled the choices in the character-specific interaction sequences as affiliation and power coordinates; if the brain represents these coordinates, participants may have subjective access to them. Thus, we expected participants' post-task placements of the characters into a 2D affiliation and power space to be closer to their end-of-task behavioral locations than expected by chance. For each participant in the Validation sample, we calculated the mapping error as the average Euclidean distance between the character-wise behavioral and self-reported locations. To establish participant-specific chance distributions, we permuted the subjective locations and re-calculated this error 100 times for each participant. We compared each participant's average mapping error against their chance distribution by calculating a z-score with respect to the permutation distribution. We then tested whether these mapping errors were smaller than 0 with a left-tailed t-test.

Given that retrieving the subjective locations should be memory-dependent, we also predicted a negative relationship between mapping error and post-task memory recall: participants with better memory of the interactions should place the characters closer to their behavioral locations. We tested this prediction using Pearson's correlation and a left-tailed p-value.

fMRI acquisition and pre-statistics

Image acquisition

Scans were collected in a single run of approximately 26 minutes. Both samples' data were acquired on 3-Tesla (3T) scanners but with different image acquisition parameters.

Initial sample

Images were collected on a Siemens Allegra 3T scanner (Siemens, Erlangen, Germany). T2*-weighted images were collected with a single-shot echo-planar imaging (EPI) pulse sequence with the following parameters: flip angle = 90° , echo time (TE) = 35 milliseconds (ms), repetition time (TR) = 2 seconds (s), 36 slices, 64×64 matrix, voxel size = 3 mm^3 . T1-weighted images were collected with a magnetization-prepared rapid gradient-echo (MPRAGE) protocol with voxel size = 1 mm^3 .

Validation sample

Images were collected on a Siemens Skyra 3T scanner. T2*-weighted images were collected with a multiband slice EPI pulse sequence with the following parameters: multiband acceleration factor = 7, flip angle = 60° , TE = 35 ms, TR = 1 s, 70 slices, 108×108 matrix, voxel size = 2.1 mm^3 . T1-weighted images were collected with a MPRAGE protocol with voxel size = 0.8 mm^3 .

Image preprocessing

Image preprocessing was conducted using SPM12 (Wellcome Trust Centre for Neuroimaging), with standard steps. To correct for head motion, the functional images were realigned to the first volume using 6 parameter rigid body transformation (3 translations and 3 rotations) and then unwarped to account for magnetic field inhomogeneities. The realigned images were slice-time corrected to the middle slice using normalized mutual information, then co-registered in alignment with the MPRAGE to the mean unwarped image. The MPRAGE image was then segmented into 6 tissue classes (gray matter, white matter, cerebral spinal fluid, skull, soft-tissue and air) and the resulting forward deformation parameters were used to normalize the images to a standard Montreal Neurological Institute (MNI) template image (using 4th-degree B-spline interpolation). The functional images were not smoothed.

fMRI analysis

Trial-wise general linear modeling

To estimate single trial activation patterns, general linear models (GLMs) were fitted to each voxel in the functional images using SPM12. Unsmoothed images were used to preserve the spatial resolution of the multi-voxel patterns. Microtime resolution was set to the number of slices collected and microtime onset was set to the middle slice. To reduce temporal autocorrelation in the time series, we used a high-pass filter of 128 seconds (1/128 Hertz) and prewhitening (with SPM's FAST algorithm).

The design matrices had separate regressors for each decision trial to estimate all decision trials' weights in the same regression (i.e., a least-squares-all approach). For different analyses (see below), we modeled the events with a stick function at trial onset or boxcar function over the reaction time (from onset to the button press), depending on whether the neural representation was assumed to be independent of the decision. The narrative trial onsets were modeled by an additional regressor. The decision and narrative trial regressors were convolved with a canonical hemodynamic response function to model the blood oxygenation level dependent (BOLD) signal changes expected from the events. The six realignment parameters from preprocessing were included in the design matrices to regress out residual motion-related variance. To further improve the signal-to-noise ratio, the GLMs were only estimated for voxels with an average signal greater than 50% of the global signal. After GLM estimation, each participant had a beta image with a beta series ($\beta_1, \beta_2, \dots, \beta_{63}$) for each voxel that reflects trial-specific activation magnitudes. The Initial sample's beta images were resampled to match the higher spatial resolution Validation sample's images, so that all analyses cover the same number of voxels and amount of brain volume across participants.

Social dimension similarity searchlight analysis

The affiliation and power dimensions are akin to social contexts: we expected the hippocampus to represent interactions of the same dimension more similarly than interactions of different dimensions. In other words, we expected the affiliation and power interactions to be represented abstractly, such that their representations generalize across trials. We used a representational similarity analysis (RSA) searchlight to test this hypothesis (see **Searchlight analysis details** below). We assumed that representations of social dimension are independent of the decisions themselves, and so we used participant-specific trial-wise beta images where each decision trial was modeled with a stick function regressor at the trial onset. We expected the same dimension trial pairs to have more correlated neural patterns than different dimension trial pairs. For each participant and each searchlight sphere, we calculated the difference between the average neural pattern correlations (Pearson's r with Fisher's z -transform) for same dimension trial pairs versus different dimension trial pairs; the resulting statistic was stored in the center voxel.

Importantly, this analysis naturally balances various narrative-related variables. As described in the methods (see **Affiliation and power decisions**), affiliation and power trials were not different in word count or sentiment. Character identity is also balanced, because the characters appear in the same number of affiliation and power trials. Further, the number of previous decision trials

with specific characters were not statistically different between same and different dimension trial pairs (t -test $p = 0.55$). However, there was a significant difference in the average reaction time between affiliation and power decisions across participants ($t_{49} = 6.92$, $p < 0.001$; affiliation mean = 4.92 seconds (s), power mean = 4.51 s), so we controlled for this in the group-level analysis.

Location similarity searchlight analyses

We also expected the hippocampus to represent the different characters' changing social locations, which are implicit in the participant's choices. We used multiple regression searchlight RSA to test whether hippocampal pattern dissimilarity increases with social location distance, based on participant-specific trial-wise beta images where boxcar regressors spanned each trial's reaction time.

We ran two complementary regression analyses to address two related questions. First, we asked whether the hippocampus represents how a specific relationship changes over time. For this analysis, for each participant and each searchlight, we computed character-specific (i.e., only for same character trial pairs) correlation distances between trial-wise beta patterns and Euclidean distances between the social location behavioral coordinates. Distances were z-scored within character trial pairs to isolate character-specific changes. The second analysis asked whether there is a common map-like representation, where all trials, regardless of relationship, are represented in a shared coordinate system. Here, we included all trial pairs and z-scored the distances globally.

For both regression analyses, we included control distances to control for possible confounds. To account for generic time-related changes, we controlled for absolute scan-time difference, as this correlated with location distance across participants (see **Temporal autocorrelation of hippocampal beta patterns** in the supplement). Although the square of this temporal distance did not explain any additional variance in behavioral distances, we ran a robustness analysis including both temporal distance and its square and saw qualitatively the same clusters with similar effect sizes. As such, we report the main analysis only. We included binary dimension difference (0 = trial pairs of different dimension, 1 = trials pairs of the same dimension), to ensure effects could not be explained by dimension-related effects. In the group-level model, we controlled for sample and the average reaction time between affiliation and power decisions.

Searchlight analysis details

Each searchlight was a sphere with a diameter of 11 voxels. We smoothed the participant-level searchlight images with an 8mm Full Width at Half Maximum (FWHM) kernel and used a group-level voxel-wise group-level General Linear Model, controlling for sample and the average reaction time difference between affiliation and power decisions. We used right-tail tests to test whether the hypothesized effect had a positive relationship to neural pattern similarity. The family-wise error rate (FWER) for significant testing was controlled at an alpha of 0.05, using a max-type permutation testing procedure²⁹.

Region-of-interest details

In our searchlight analyses, we specifically tested the bilateral hippocampus^{6,10}. We defined left and right hippocampal masks from the Harvard-Oxford Atlas, co-registered and re-sampled them to the Validation sample's functional images and then binarized at a 50% probability threshold. We used small volume correction in the hippocampi for our main tests. For whole-brain exploratory analyses, we corrected for the total number of (non-hippocampal) comparisons, and thresholded so that clusters were at least 50 contiguous voxels.

Data availability

The data are available at: <https://osf.io/xqgr3/>²⁹. The code for the analyses are available at: <https://github.com/matty-gee/SocialTrajectories>²⁹.

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Additional information

Author contributions

MS and DS conceived of the study. PKB, VS and KB collected the data. MS ran the analyses. MS, DS and KB wrote the manuscript.

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Additional files

[Supplemental file](#) 

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Peer reviews

Reviewer #1 (Public review):

Schafer et al. tested whether the hippocampus tracks social interactions as sequences of neural states within an abstract social space defined by the dimensions of affiliation and power, using a narrative-based task in which participants engaged in dynamic social interactions. The study showed that individual social relationships were represented as distinct trajectories of hippocampal activity patterns. These neural trajectories systematically reflected trial-by-trial changes in affiliation and power between the participant and each character, suggesting that the hippocampus encodes sequences of socially relevant events and their relational structure, extending its well-established role beyond spatial representations.

A major strength of this study is the use of a richly structured, narrative-based task that allows social relationships to evolve dynamically over time. The use of representational similarity analysis provides a principled framework for linking behavioral trajectories in social space to neural pattern dynamics.

One potential limitation concerns temporal autocorrelation in the neural data, as nearby trials are inherently related both behaviorally and temporally within a continuous narrative. Although the authors carefully attempted to control for temporal distance and related confounds, fully disentangling representational similarity driven by social structure from similarity driven by temporal proximity remains challenging within a single-session task design.

While the findings of a two-dimensional representational structure is an important contribution, it remains an open question whether such a representation reflects an inherent property of how the human brain encodes social relationships, or whether it is partly driven by task constraints in which social interactions were limited to changes along two (affiliation and power) dimensions. Future studies that allow social relationships to vary along richer or higher-dimensional feature spaces will be necessary to determine the generality of low dimensional representations.

<https://doi.org/10.7554/eLife.96895.2.sa2>

Reviewer #2 (Public review):

The substantially revised paper has increased in clarity and is much more accessible and straightforward than the first version. The analyses are now clearer and support the conclusions better. There are however some remaining methodological weaknesses, which in my mind still renders the evidence to not be entirely convincing.

(1) The temporal autocorrelation concern is not fully convincingly addressed. The temporal autocorrelation curves supplied in the supplements are really helpful, but linearly regressing out the temporal distance from the neural distance clearly does not work, as one can see from the right panel of supplementary Figure 1. If the method had worked correctly the line should have been flat. The analysis however shows that decision trials with a lag > 2 are basically independent - so a simple way to address this is to restrict the RSA analysis to trials with a decision lag of > 2. This analysis would strengthen the paper a lot.

(2) In the final analysis, the authors use all the trials to make the claim that the hippocampus represents the characters in a shared social space. However, as within-character distances are still included in the analysis, this result could still be driven by the effects of within-character representations that are not shared across characters. A simple way of addressing

this concern would be to only include between-character distances in this analysis, making it truly complementary to the previous within-character analysis. It would also be very interesting to compare the within- and between-character analyses in the hippocampus directly.

(3) Overall, the correction for multiple comparisons in the fMRI and the resulting corrected p-values are not sufficiently explained and documented in the paper. What was exactly permuted in the tests? Was correction applied in a voxel-wise or cluster-wise fashion? If cluster-wise, the cluster-wise p-values need to be reported.

<https://doi.org/10.7554/eLife.96895.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Public reviews:

Reviewer #1 (Public review):

Summary:

Schafer et al. tested whether the hippocampus tracks social interactions as sequences of neural states within an abstract social space defined by dimensions of affiliation and power, using a task in which participants engaged in narrative-based social interactions. The findings of this study revealed that individual social relationships are represented by unique sequences of hippocampal activity patterns. These neural trajectories corresponded to the history of trial-to-trial affiliation and power dynamics between participants and each character, suggesting an extended role of the hippocampus in encoding sequences of events beyond spatial relationships.

The current version has limited information on details in decoding and clustering analyses which can be improved in the future revision.

Strengths:

(1) Robust Analysis: The research combined representational similarity analysis with manifold analyses, enhancing the robustness of the findings and the interpretation of the hippocampus's role in social cognition.

(2) Replicability: The study included two independent samples, which strengthens the generalizability and reliability of the results.

Weaknesses:

I appreciate the authors for utilizing contemporary machine-learning techniques to analyze neuroimaging data and examine the intricacies of human cognition. However, the manuscript would benefit from a more detailed explanation of the rationale behind the selection of each method and a thorough description of the validation procedures. Such clarifications are essential to understand the true impact of the research. Moreover, refining these areas will broaden the manuscript's accessibility to a diverse audience.

We thank the reviewer for these comments and have addressed them in various ways.

First, we removed the spline-based decoding and spectral clustering analyses. As we detail in our response to the recommendations, these approaches were complex and raised legitimate interpretational concerns, making it unclear how they supported our core claims. The revised manuscript now focuses on a set of representational similarity analyses to show

representations consistent with social dimension similarity (affiliation vs. power decision trials) and social location similarity (trajectory/map-like coding based on participant choices).

Second, we expanded the Methods and Results to more clearly explain the analyses, the questions they address, and associated controls and robustness tests. The dimension similarity analysis tests whether hippocampal patterns differentiate affiliation and power decisions in a way consistent with an abstract dimension representation. The location similarity RSAs test whether within-character neural pattern distances scale with Euclidean distance in social space (relationship-specific trajectories), and whether pattern distances across all characters scale with location distances when distances are globally standardized, consistent with a shared map-like coordinate system.

Third, we emphasize new controls. For the dimension similarity RSA, we test for potential confounds such as word count, text sentiment, and reaction time differences between affiliation and power trials. For the location similarity RSA, we control for temporal distance between trials and show (in the Supplement) that the reported effects cannot be explained by temporal autocorrelation in the fMRI data or by the relationship between temporal distance and behavioral location distance.

We believe that these changes address the reviewer's request for clearer rationale and validation.

Reviewer #2 (Public review):

Summary:

Using an innovative task design and analysis approach, the authors set out to show that the activity patterns in the hippocampus related to the development of social relationships with multiple partners in a virtual game. While I found the paper highly interesting (and would be thrilled if the claims made in the paper turned out to be true), I found many of the analyses presented either unconvincing or slightly unconnected to the claims that they were supposed to support. I very much hope the authors can alleviate these concerns in a revision of the paper.

Strengths & Weaknesses:

(1) The innovative task design and analyses, and the two independent samples of participants are clear strengths of the paper.

We thank the reviewer for this comment.

(2) The RSA analysis is not what I expected after I read the abstract and title of the result section "The hippocampus represents abstract dimensions of affiliation and power". To me, the title suggests that the hippocampus has voxel patterns, which could be read out by a downstream area to infer the affiliation and power value, independent of the exact identity of the character in the current trial. The presented RSA analysis however presents something entirely different - namely that the affiliation trials and power trials elicit different activity patterns in the area indicated in Figure 3. What is the meaning of this analysis? It is not clear to me what is being "decoded" here and alternative explanations have not been considered. How do affiliation and power trials differ in terms of the length of sentences, complexity of the statements, and reaction time? Can the subsequent decision be decoded from these areas? I hope in the revision the authors can test these ideas - and also explain how the current RSA analysis relates to a representation of the "dimensions of affiliation and power".

We agree that this analysis needed to be better justified and explained. We have revised the text to clarify that by "represents the interaction decision trials along abstract social

dimensions” we mean that hippocampal multivoxel patterns differentiate affiliation and power decisions in a way consistent with the conceptual framework of underlying latent dimensions. The analysis tests one simple prediction of this view – that on average these trial types are separable in the neural patterns. We have added details to the Methods, showing how the affiliation and power trials do not differ in word count or in sentiment, but do differ in their semantics, as assessed by a Large Language Model, as we expect from our task assumptions. Thanks to the reviewer’s comment, we also tested for and found a reaction time difference between affiliation and power trials, that we now control for.

(3) Overall, I found that the paper was missing some more fundamental and simpler RSA analyses that would provide a necessary backdrop for the more complicated analyses that followed. Can you decode character identity from the regions in question? If you trained a simple decoder for power and affiliation values (using the LLE, but without consideration of the sequential position as used in the spline analysis), could you predict left-out trials? Are affiliation and power represented in a way that is consistent across participants - i.e. could you train a model that predicts affiliation and power from N-1 subjects and then predict the Nth subject? Even if the answer to these questions is "no", I believe that they are important to report for the reader to get a full understanding of the nature of the neural representations in these areas. If the claim is that the hippocampus represents an "abstract" relationship space, then I think it is important to show that these representations hold across relationships. Otherwise, the claim needs to be adjusted to say that it is a representation of a relationship-specific trajectory, but not an abstract social space.

We appreciate this comment and agree on the value of clear, conceptually simple analyses. To address this concern, we have simplified our main analysis significantly by removing the spline-based analysis and substituting it with a multiple regression representational similarity analysis approach. We test whether within-character neural pattern distances scale with distance in social space (relationship-specific trajectories), and whether pattern distances across all characters scale with location distances when distances are globally standardized. We find evidence for both, consistent with a shared map-like coordinate system.

We agree that decoding character identity and an across-participant decoding approach could be informative. However, our current task is not well designed for such analyses and as such would complicate the paper. Although we agree that these questions are interesting, they would test questions that are outside the scope of this paper.

(4) To determine that the location of a specific character can be decoded from the hippocampal activity patterns, the authors use a sequential analysis in a lowdimensional space (using local linear embedding). In essence, each trial is decoded by finding the pair of two temporally sequential trials that is closest to this pattern, and then interpolating the power/affiliation values linearly between these two points. The obvious problem with this analysis is that fMRI pattern will have temporal autocorrelation and the power and affiliation values have temporal autocorrelation. Successful decoding could just reflect this smoothness in both time series. The authors present a series of control analyses, but I found most of them to not be incisive or convincing and I believe that they (and their explanation of their rationale) need to be improved. For example, the circular shifting of the patterns preserves some of the autocorrelation of the time series - but not entirely. In the shifted patterns, the first and last items are considered to be neighboring and used in the evaluation, which alone could explain the poor performance. The simplest way that I can see is to also connect the first and last item in a circular fashion, even when evaluating the veridical ordering. The only really convincing control condition I found was the generation of new sequences for every character by shuffling the sequence of choices and re-creating new artificial trajectories with the same start and endpoint. This

analysis performs much better than chance (circular shuffling), suggesting to me that a lot of the observed decoding accuracy is indeed simply caused by the temporal smoothness of both time series.

We thank the reviewer for emphasizing this important concern; we agree that we did not sufficiently address this in the initial submission. This concern is one main reason we removed the spline-based analysis and now use regression-based representational similarity analyses in its place. In the revision, we report autocorrelation-related analyses in the supplement, and via controls and additional analysis show that temporal distance (or its square) cannot explain the location-like effects. This substantially improves our ability to interpret the findings.

(5) Overall, I found the analysis of the brain-behavior correlation presented in Figure 5 unconvincing. First, the correlation is mostly driven by one individual with a large network size and a 6.5 cluster. I suspect that the exclusion of this individual would lead to the correlation losing significance. Secondly, the neural measure used for this analysis (determining the number of optimal clusters that maximize the overlap between neural clustering and behavioral clustering) is new, non-validated, and disconnected from all the analyses that had been reported previously. The authors need to forgive me for saying so, but at this point of the paper, would it not be much more obvious to use the decoding accuracy for power and affiliation from the main model used in the paper thus far? Does this correlate? Another obvious candidate would be the decoding accuracy for character identity or the size of the region that encodes affiliation and power. Given the plethora of candidate neural measures, I would appreciate if the authors reported the other neural measures that were tried (and that did not correlate). One way to address this would have been to select the method on the initial sample and then test it on the validation sample - unfortunately, the measure was not pre-registered before the validation sample was collected. It seems that the correlation was only found and reported on the validation sample?

We agree that this analysis was too complicated and under constrained, and thus not convincing. We think that removing this cluster-based analysis is the most conservative response to the reviewer's concerns and have removed it from the revised paper.

Recommendations to the authors:

Reviewer #1 (Recommendations for the authors):

The manuscript's description of the shuffling analysis performed during decoding is currently ambiguous, particularly concerning the control variables. This ambiguity is present only in the Figure 4 legends and requires a more detailed explanation within the methods section. It is essential to clarify whether the permutation process was conducted within each character's data set or across multiple characters' data sets. If permutations were confined to within-character data, the conclusion would be that the hippocampus encodes context-specific information rather than providing a twodimensional common space.

We thank the reviewer for this comment. We have now removed the spline analysis due to these and other problems and have replaced it with representational similarity analyses that are both more rigorous and easier to interpret. We think these analyses allow us to make the claim that the characters are represented in a common space.

In the methods, we explain the analyses (page 23-24, lines 475-500):

“We also expected the hippocampus to represent the different characters' changing social locations, which are implicit in the participant's choices. We used multiple regression searchlight RSA to test whether hippocampal pattern dissimilarity increases with social

location distance, based on participant-specific trial-wise beta images where boxcar regressors spanned each trial's reaction time."

"We ran two complementary regression analyses to address two related questions. First, we asked whether the hippocampus represents how a specific relationship changes over time. For this analysis, for each participant and each searchlight, we computed character-specific (i.e., only for same character trial pairs) correlation distances between trial-wise beta patterns and Euclidean distances between the social location behavioral coordinates. Distances were z-scored within character trial pairs to isolate character-specific changes. The second analysis asked whether there is a common map-like representation, where all trials, regardless of relationship, are represented in a shared coordinate system. Here, we included all trial pairs and z-scored the distances globally. For both regression analyses, we included control distances to control for possible confounds. To account for generic time-related changes, we controlled for absolute scan-time difference, as this correlated with location distance across participants (see Temporal autocorrelation of hippocampal beta patterns in the supplement). Although the square of this temporal distance did not explain any additional variance in behavioral distances, we ran a robustness analysis including both temporal distance and its square and saw qualitatively the same clusters with similar effect sizes. As such, we report the main analysis only. We included binary dimension difference (0 = trial pairs of different dimension, 1 = trials pairs of the same dimension), to ensure effects could not be explained by dimension-related effects. In the group-level model, we controlled for sample and the average reaction time between affiliation and power decisions."

In the results, we describe the results and our interpretation (pages 11-12, lines 185208):

"We have shown that the left hippocampus represents the affiliation and power trials differently, consistent with an abstract dimensional representation. Does it also represent the changing social coordinates of each character? To test this, we multiple-regression RSA searchlight to test whether left hippocampus patterns represent the characters' changing social locations across interactions (see Figure 3). We restricted the distances to those from trial pairs from the same character and standardized the distances within character (see Figure 3BD). We controlled for temporal distance to ensure the effect was not explainable by the time between trials, and for whether the trials shared the same underlying dimension (affiliation or power; see Location similarity searchlight analyses for more details). At the group level, we controlled for sample and the average reaction time difference between affiliation and power trials. Using the same testing logic as the dimensionality similarity analysis, we first tested our hypothesis in the bilateral hippocampus and found widespread effects in both the left (peak voxel MNI x/y/z = -35/-22/-15, cluster extent = 1470 voxels) and right (peak voxel MNI x/y/z = 37/-19/-14, cluster extent = 1953 voxels) hemispheres. The whole-brain searchlight analysis revealed additional clusters in the left putamen (-27/-3/14, cluster extent = 131 voxels) and left posterior cingulate cortex (-10/-28/41, cluster extent = 304 voxels)."

"We then asked a second, complementary question: does the hippocampus represent all interactions, across characters, within a shared map? To test for this map-like structure, we repeated the analysis but now included all trial pairs, z-scoring distances globally rather than within character (Figure 3E-F). The remainder of the procedure followed the same logic as the preceding analysis. The hippocampus analysis revealed an extensive right hippocampal cluster (27/27/-14, cluster extent = 1667 voxels). The whole-brain analysis did not show any significant clusters."

We also describe the results in the discussion (page 12, lines 220-226):

"Then, we show that the hippocampus tracks the changing social locations (affiliation and power coordinates), above and beyond the effects of dimension or time; the hippocampus seemed to reflect both the changing within-character locations, tracking their locations over

time, and locations across characters, as if in a shared map. Thus, these results suggest that the hippocampus does not just encode static character-related representations but rather tracks relationship changes in terms of underlying affiliation and power.”

The manuscript's description of the decoding analysis is unclear regarding the variability of the decoded positions. The authors appear to decode the position of a character along a spline, which raises the question of whether this position correlates with time, since characters are more likely to be located further from the center in later trials. There is a concern that the decoded position may not solely reflect the hippocampal encoding of spatial location, but could also be influenced by an inherent temporal association. Given that a character's position at time t is likely to be similar to its positions at $t-1$ and $t+1$, it is crucial that the authors clearly articulate their approach to separating spatial representation from temporal autocorrelation. While this issue may have been addressed in the construction of the test set, the manuscript does not seem to adequately explain how such biases were mitigated in the training set.

We agree that temporal confounding needs to be better accounted for, as our claims depend on space-like signals being separable from time-like ones. We address this in several ways in the revised manuscript.

First, we emphasize that this is a narrative-based task, where temporal structure is relevant. As such, our analyses aim to demonstrate that effects go beyond simple temporal confounds, like trial order or time elapsed.

Despite the temporal structure to the task, the decisions for the same character are spaced in time, and interleaved with other characters' decisions, reducing the chance that a simple temporal confound could explain trajectory-related effects. We now describe the task better in the revised methods (page 16, lines 314-318):

“All six characters' decision trials are interleaved with one another and with narrative slides. On average, after a decision trial for a given character, participants view ~11 narrative slides and complete ~3 decisions for other characters before returning to that same character, such that each character's choices are separated by an average of ~20 seconds (range 12 seconds to 10 min).”

To address temporal autocorrelation in the fMRI time series, we used SPM's FAST algorithm. Briefly, FAST models temporal autocorrelation as a weighted combination of candidate correlation functions, using the best estimate to remove autocorrelated signal.

We also now report the temporal autocorrelation profile of the hippocampal beta series in the supplement, including (pages 29-31, lines 593-656):

“The Social Navigation Task is a narrative-based task, where the relationships with characters evolve over time; trial pairs that are close in time may have more similar fMRI patterns for reasons unrelated to social mapping (e.g., slow drift). It is important to account for the role of time in our analyses, to ensure effects go beyond simple temporal confounds, like the time between decision trials. To aid in this, we quantified how fMRI signals change over time using a pattern autocorrelation function across decision trial lags. We defined the left and right hippocampus and the left and right intracalcarine cortex using the HarvardOxford atlas and thresholded them at 50% probability. We chose intracalcarine cortex as an early visual control region that largely corresponds to primary visual cortex (V1), as it is likely to be driven by the visually presented narrative. We used the same trial-wise beta images as in the location similarity RSA (boxcar regressors spanning each decision trial's reaction time). For each participant and region-of-interest (ROI), we extracted the decision trial-by-voxel beta matrix and quantified three kinds of temporal dependence: beta autocorrelation, multivoxel pattern correlation and multivoxel pattern correlation after regressing out temporal distance.”

“To estimate the temporal autocorrelation of the trial-wise beta values, we treated each voxel’s beta values as a time series across trials and measured how much a voxel’s response on one trial correlated (Pearson) with its response on previous trials. We averaged these voxel wise autocorrelations within each ROI. At one trial apart (lag 1), both the hippocampus and V1 showed small positive autocorrelations, indicating modest trial-to-trial carryover in response amplitude (see Supplemental figure 1) that by three trials apart was approximately 0.”

“Because our representational similarity analyses depend on trial-by-trial pattern similarity, we also estimated how multivoxel patterns were autocorrelated over time. For each lag, we computed the Pearson correlation between each trial’s voxelwise pattern and the pattern from the trial that many trials earlier, then averaged those correlations to obtain a single autocorrelation value for that lag. At one trial apart, both regions showed positive autocorrelation, with V1 having greater autocorrelation than the hippocampus; pattern correlations between trials 3 or 4 trials apart reduced across participants, settling into low but positive values. Then, for each participant and ROI, we regressed out the effect of absolute trial onset differences from all pairwise pattern correlations, to mirror the effects of controlling for these temporal distances in regressions. After removing this temporal distance component, the short lag pattern autocorrelation dropped substantially in both regions. The similarity in autocorrelation profiles between the two regions suggests that significant similarity effects in the hippocampus are unlikely to be driven by generic temporal autocorrelation.”

“Relationship between behavioral location distance and temporal distance “

“We also quantified how temporal distances between trials relates to their behavioral location distances, participant by participant. Our dimension similarity analysis controls for temporal distance between trials by design (see Social dimension similarity searchlight analysis), but our location similarity analysis does not. To decide on covariates to include in the analysis, we tested whether temporal distances can explain behavioral location distances. For each participant, we computed the correlations between trial pairs’ Euclidean distances in social locations and their linear temporal distances (“linear”) and the temporal distances squared (“quadratic”), to test for nonlinear effects. We then summarized the correlations using one-sample t-tests. The linear relationship was statistically significant ($t_{49} = 12.24$, $p < 0.001$), whereas the quadratic relationship was not ($t_{49} = -0.55$, $p = 0.586$). Similarly, in participant specific regressions with both linear and quadratic temporal distances, the linear effect was significant ($t_{49} = 5.69$, $p < 0.001$) whereas the quadratic effect was not ($t_{49} = 0.20$, $p = 0.84$). Based on this, we included linear temporal distances as a covariate in our location similarity analyses (see Location similarity searchlight analyses), and verified that adding a quadratic temporal distance covariate does not alter the results. Thus, the reported location-related pattern similarity effects go beyond what can be explained by temporal distance alone.”

How the free parameter of spectral clustering was determined, if there is any?

The interpretation of the number of hippocampal activity clusters is ambiguous. It is suggested that this number could fluctuate due to unique activity patterns or the fit to behaviorally defined trajectories. A lower number of clusters might indicate either a noisier or less distinct representation, raising the question of the necessity and interpretability of such a complex analysis. This concern is compounded by the potential sensitivity of the clustering to the variance in Euclidean distances of each trial's position relative to the center. If a character's position is consistently near the center, this could artificially reduce the perceived number of clusters. Furthermore, the manuscript should address whether there is any correlation between the number of clusters and behavioral performance. Specifically, what are the implications if participants are able to perform

the task adequately with a smaller number of distinct hippocampal representation states?

The rationale for conducting both cluster analysis and position decoding as separate analyses remains unclear. While cluster analysis can corroborate the findings of position decoding, it is not apparent why the authors chose to include trials across characters for cluster analysis but not for decoding analysis. An explanation of the reasoning behind this methodological divergence would help in understanding the distinct contributions of each analysis to the study's findings.

The paper by Cohen et al. (1997), which provides the questionnaire for measuring the social network index, is not cited in the references. Upon reviewing the questionnaire that the author may have used, it appears that the term "social network size" does not refer to the actual size but to a score or index derived from the questionnaire responses. It may be more appropriate to replace the term "size" with a different term to more accurately reflect this distinction.

Thank you for seeking these clarifications. Given the complexity of this analysis, we have decided to drop it to focus instead on our dimension and location representational similarity analysis results.

Reviewer #2 (Recommendations for the authors):

How did the participants' decisions on previous trials influence the future trials that the subjects saw? If the different participants were faced with different decision trials, then how did you compare their decision? If two participants made the same decisions, would they have seen exactly the same sequence of trials (see point X on how the trial sequence was randomized).

All participants experience the same narrative, with the same decisions (i.e., the same available options); their choices (i.e., the options they select) are what implicitly shape each character's affiliation and power locations, and thus each character's trajectory. In other words, the narrative is fixed; what changes is the social coordinates assigned to each trial's outcome depending on the participant's choice of how to interact from the two narrative options. This means that we can meaningfully compare participants' neural patterns, given that every participant received the same text and images throughout.

We have now added details on the narrative structure, replacing more ambiguous statements with a clearer description (page 16, lines 309-318):

"The sequence of trials, including both narrative and decision trials, were fixed across participants; all that differs are the choices that the participants make. Narrative trials varied in duration, depending on the content (range 2-10 seconds), but were identical across participants. Decision trials always lasted 12 seconds, with two options presented until the participant made a choice, after which a blank screen was presented for the remainder of the duration. All six characters' decision trials are interleaved with one another, and with the narrative slides. On average, after a decision trial for a given character, participants view ~11 narrative slides and complete ~3 decisions for other characters before returning to another decision with the same character, such that each character's choices are separated by an average of ~20 seconds (ranging from 12 seconds to 10 min)."

Figure 2B: I assume that "count" is "count of participants"? It would be good to indicate this on the axis/caption.

Thank you for noting this. We have now removed this figure to improve the clarity of our figures.

We have shown that the hippocampus represents the interaction decision trials along abstract social dimensions, but does it track each relationship's unique sequence of abstract social coordinates?". Please clarify what you mean by "represents the interaction decision trials".

By “represents the interaction decision trials along abstract social dimensions”, we mean that when the participant makes a choice during the social interactions the hippocampal patterns represent the current social dimension of the choice (affiliation vs power). In other words, the hippocampal BOLD patterns differentiate affiliation and power decisions, consistent with our hypothesis of abstract social dimension representation in the hippocampus. We have clarified this (page 11, lines 185-187):

“We have shown that the left hippocampus represents the affiliation and power trials differently, consistent with an abstract dimensional representation.”

Page 8: "Hippocampal sequences are ordered like trajectories": It is not entirely clear to me what is meant by the split midpoint. Is this the midpoint of the piece-wise linear interpolation between two points, or simply the mean of all piecewise splines from one character? If the latter, is the null model the same as simply predicting the mean affiliation and power value for this character? If yes, please clarify and simplify this for the reader.

Page 8: "Hippocampal sequences track relationship-specific paths". First, I was misled by the "relationship-specific". I first understood this to mean that you wanted to test whether two relationships (i.e. the identity of the partner) had different representations in Hippocampus, even if the power/affiliation trajectories are the same. I suggest changing the title of this section.

The analysis in this section also breaks any temporal autocorrelation of measured patterns - so I am not sure if this is a strong analysis that should be interpreted at all. This analysis seems to not address the claim and conclusion that is drawn from it. I assume that the random trajectories have different choices and different affiliation/power values than the true trajectories. So the fact that the true trajectories can be better decoded simply shows that either choices or affiliation and power (or both) are represented in the neural code - but not necessarily anything beyond this.

Page 9: "Neural trajectories reflect social locations, not just choices". The motivation of this analysis is not clear to me. As I understand this analysis, both social location and choices are changed from the real trajectories. How can it then show that it reflects social locations, not just the choices?

Figure 4 caption: "on the -based approximation" Is there a missing "point"-[based] here?

We agree with the reviewer that this analysis is hard to interpret and does not adequately address concerns regarding temporal autocorrelation, and as such we have removed it from the manuscript. We describe the new results that include controlling for temporal distance between trials (pages 11-12, lines 185-208):

“We have shown that the left hippocampus represents the affiliation and power trials differently, consistent with an abstract dimensional representation. Does it also represent the changing social coordinates of each character? To test this, we multiple-regression RSA searchlight to test whether left hippocampus patterns represent the characters’ changing social locations across interactions (see Figure 3). We restricted the distances to those from trial pairs from the same character and standardized the distances within character (see Figure 3BD). We controlled for temporal distance to ensure the effect was not explainable by the time between trials, and for whether the trials shared the same underlying dimension

(affiliation or power; see Location similarity searchlight analyses for more details). At the group level, we controlled for sample and the average reaction time difference between affiliation and power trials. Using the same testing logic as the dimensionality similarity analysis, we first tested our hypothesis in the bilateral hippocampus and found widespread effects in both the left (peak voxel MNI x/y/z = -35/-22/-15, cluster extent = 1470 voxels) and right (peak voxel MNI x/y/z = 37/-19/-14, cluster extent = 1953 voxels) hemispheres. The whole-brain searchlight analysis revealed additional clusters in the left putamen (-27/-3/14, cluster extent = 131 voxels) and left posterior cingulate cortex (-10/-28/41, cluster extent = 304 voxels)."

"We then asked a second, complementary question: does the hippocampus represent all interactions, across characters, within a shared map? To test for this map-like structure, we repeated the analysis but now included all trial pairs, z-scoring distances globally rather than within character (Figure 3E-F). The remainder of the procedure followed the same logic as the preceding analysis. The hippocampus analysis revealed an extensive right hippocampal cluster (27/27/-14, cluster extent = 1667 voxels). The whole-brain analysis did not show any significant clusters."

We emphasize that the results are robust to the inclusion of temporal distance squared, in the methods (pages 23-24, lines 493-496):

"Although the square of this temporal distance did not explain any additional variance in behavioral distances, we ran a robustness analysis including both temporal distance and its square and saw qualitatively the same clusters with similar effect sizes."

Page 8: last paragraph: The text sounds like you have already shown that you can decode character identity from the patterns - but I do not believe you have it this point. I would consider this would be an interesting addition to the paper, though.

This section has been removed, and we have been careful to not imply this in the current version of the manuscript. While we agree a character identity decoding would enrich our argument, we do not believe our task is well-suited to capture a character identity effect. Each character only has 12 decision trials, and these trials are partially clustered in time - this is one problem of temporal autocorrelation that we thank the reviewers for pushing us to consider in more detail. Dimension and location patterns, on the other hand, are more natural to analyze in our task, especially in representational similarity analyses that test whether the relevant differences scale with neural distances.

Page 14ff: Why is "Analysis section" not part of "Materials and Methods"? I believe adding the analysis after a careful description of the methods would improve the clarity of this section.

We agree with the reviewer and have now consolidated these two sections.

Two or three examples of Affiliation and Power decision trials should be provided, so the reader can form a more thorough understanding of how these dimensions were operationalized. For the RSA analysis, it is important to consider other differences between these two types of trials.

We agree that adding examples will clarify the operationalization of these dimensions. We now include example affiliation and power trials in a table (page 17-18).

We thank the reviewer for noting the need to rule out alternative hypotheses; we have added several such tests. Affiliation and power trials were not different in word count (page 17, lines 329-332):

“To ensure that any observed neural or behavioral differences were not confounded by trivial features of the text, we tested for differences between the affiliation and power trials (where the two options are concatenated). There were no differences in word count (affiliation average = 26.6, power average = 25.6; t-test $p = 0.56$).”

They were also not different in their sentiment, as assessed by a Large Language Model (LLM) analysis (page 17, lines 332-335):

“The text’s sentiment also did not differ between these trial types (t-test $p = 0.72$), as quantified by comparing sentiment compound scores (from most negative, -1, to most positive, +1), using a Large Language Model (LLM) specialized for sentiment analysis [26]. “

The affiliation and power trials were different in terms of semantic content, consistent with our assumptions (page 17, lines 337-347):

“Our framework assumes that affiliation and power trials differ in their semantic content—that is, in the conceptual meaning of the text, beyond word count or sentiment. To test this assumption, we used an LLM-based semantic embedding analysis. Each decision trial was embedded into a semantic vector. We then measured the cosine similarity between pairs of trials and calculated the difference between average within-dimension similarity (affiliation-affiliation and power-power comparisons) and average between-dimension similarity (affiliationpower comparisons) and assessed its statistical significance with permutation testing (1,000 shuffles of trial labels). As expected, decision trials of the same dimension were more similar to each other than trials of different dimension, across multiple LLMs (OpenAI’s text-embedding-3-small [27]: similarity difference = 0.041, $p < 0.001$; all-MiniLM-L12-v2 [28]: similarity difference = 0.032, $p < 0.001$).”

The affiliation and power trials were different in average reaction time. To control for this difference in the dimension RSA analysis, we added each participant’s absolute value reaction time difference between the trial types as a covariate. The results were nearly identical to what they were before. We updated the text to reflect this new control (page 23, lines 471-474):

“However, there was a significant difference in the average reaction time between affiliation and power decisions across participants ($t_{49} = 6.92$, $p < 0.001$; affiliation mean = 4.92 seconds (s), power mean = 4.51 s), so we controlled for this in the group-level analysis.”

The exact implementation and timing of the behavioral tasks should be described better. How many narrative trials were intermixed with the decision trials? Which characters were they assigned to? How was the sequence of trials determined? Was it fixed across participants, or randomized?

We agree that additional details are helpful. In the Methods, we now describe this with more detail (page 16, lines 301-318):

“There are two types of trials: “narrative” trials where background information is provided or characters talk or take actions (a total of 154 trials), and “decision” trials where the participant makes decisions in one-on-one interactions with a character that can change the relationship with that character (a total of 63 trials). On each decision, participants used a button response box to select between the two options. The options (1 or 2, assigned to the index and middle fingers) choice directions (+/-1 arbitrary unit on the current dimension) were counterbalanced.”

“The sequence of trials, including both narrative and decision trials, were fixed across participants; all that differs are the choices that the participants make. Narrative trials varied in duration, depending on the content (range 2-10 seconds), but were identical across

participants. Decision trials always lasted 12 seconds, with two options presented until the participant made a choice, after which a blank screen was presented for the remainder of the duration. All six characters' decision trials are interleaved with one another, and with the narrative slides. On average, after a decision trial for a given character, participants view ~11 narrative slides and complete ~3 decisions for other characters before returning to another decision with the same character, such that each character's choices are separated by an average of ~20 seconds (ranging from 12 seconds to 10 min)."

What is the exact timing of trials during fMRI acquisition - i.e. how long were the trials, what was the ITI, were there long phases of rest to determine the resting baseline? These are all important factors that will determine the covariance between regressors and should be reported carefully. Ideally, I would like to see the trial-by-trial temporal autocorrelation structure across beta-weights to be reported.

We thank the reviewer for asking for this clarification. We have added the following text to clarify the trial timing (page 16, lines 314-318):

"All six characters' decision trials are interleaved with one another and with narrative slides. On average, after a decision trial for a given character, participants view ~11 narrative slides and complete ~3 decisions for other characters before returning to that same character, such that each character's choices are separated by an average of ~20 seconds (range 12 seconds to 10 min)."

We now describe the temporal autocorrelation patterns in the supplement, including how we decided on how to control for temporal distance in representational similarity analyses (pages 29-31, lines 593-656):

"The Social Navigation Task is a narrative-based task, where the relationships with characters evolve over time; trial pairs that are close in time may have more similar fMRI patterns for reasons unrelated to social mapping (e.g., slow drift). It is important to account for the role of time in our analyses, to ensure effects go beyond simple temporal confounds, like the time between decision trials. To aid in this, we quantified how fMRI signals change over time using a pattern autocorrelation function across decision trial lags. We defined the left and right hippocampus and the left and right intracalcarine cortex using the HarvardOxford atlas and thresholded them at 50% probability. We chose intracalcarine cortex as an early visual control region that largely corresponds to primary visual cortex (V1), as it is likely to be driven by the visually presented narrative. We used the same trial-wise beta images as in the location similarity RSA (boxcar regressors spanning each decision trial's reaction time). For each participant and region-of-interest (ROI), we extracted the decision trial-by-voxel beta matrix and quantified three kinds of temporal dependence: beta autocorrelation, multivoxel pattern correlation and multivoxel pattern correlation after regressing out temporal distance."

"To estimate the temporal autocorrelation of the trial-wise beta values, we treated each voxel's beta values as a time series across trials and measured how much a voxel's response on one trial correlated (Pearson) with its response on previous trials. We averaged these voxel wise autocorrelations within each ROI. At one trial apart (lag 1), both the hippocampus and V1 showed small positive autocorrelations, indicating modest trial-to-trial carryover in response amplitude (see Supplemental figure 1) that by three trials apart was approximately 0."

"Because our representational similarity analyses depend on trial-by-trial pattern similarity, we also estimated how multivoxel patterns were autocorrelated over time. For each lag, we computed the Pearson correlation between each trial's voxelwise pattern and the pattern from the trial that many trials earlier, then averaged those correlations to obtain a single autocorrelation value for that lag. At one trial apart, both regions showed positive autocorrelation, with V1 having greater autocorrelation than the hippocampus; pattern

correlations between trials 3 or 4 trials apart reduced across participants, settling into low but positive values. Then, for each participant and ROI, we regressed out the effect of absolute trial onset differences from all pairwise pattern correlations, to mirror the effects of controlling for these temporal distances in regressions. After removing this temporal distance component, the short lag pattern autocorrelation dropped substantially in both regions. The similarity in autocorrelation profiles between the two regions suggests that significant similarity effects in the hippocampus are unlikely to be driven by generic temporal autocorrelation.”

“Relationship between behavioral location distance and temporal distance “

“We also quantified how temporal distances between trials relates to their behavioral location distances, participant by participant. Our dimension similarity analysis controls for temporal distance between trials by design (see Social dimension similarity searchlight analysis), but our location similarity analysis does not. To decide on covariates to include in the analysis, we tested whether temporal distances can explain behavioral location distances. For each participant, we computed the correlations between trial pairs’ Euclidean distances in social locations and their linear temporal distances (“linear”) and the temporal distances squared (“quadratic”), to test for nonlinear effects. We then summarized the correlations using one-sample t-tests. The linear relationship was statistically significant ($t_{49} = 12.24$, $p < 0.001$), whereas the quadratic relationship was not ($t_{49} = -0.55$, $p = 0.586$). Similarly, in participant specific regressions with both linear and quadratic temporal distances, the linear effect was significant ($t_{49} = 5.69$, $p < 0.001$) whereas the quadratic effect was not ($t_{49} = 0.20$, $p = 0.84$). Based on this, we included linear temporal distances as a covariate in our location similarity analyses (see Location similarity searchlight analyses), and verified that adding a quadratic temporal distance covariate does not alter the results. Thus, the reported location-related pattern similarity effects go beyond what can be explained by temporal distance alone.”

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